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(54) **DOPED ALUMINUM OXIDE BONDING LAYER FOR HYBRID BONDING**

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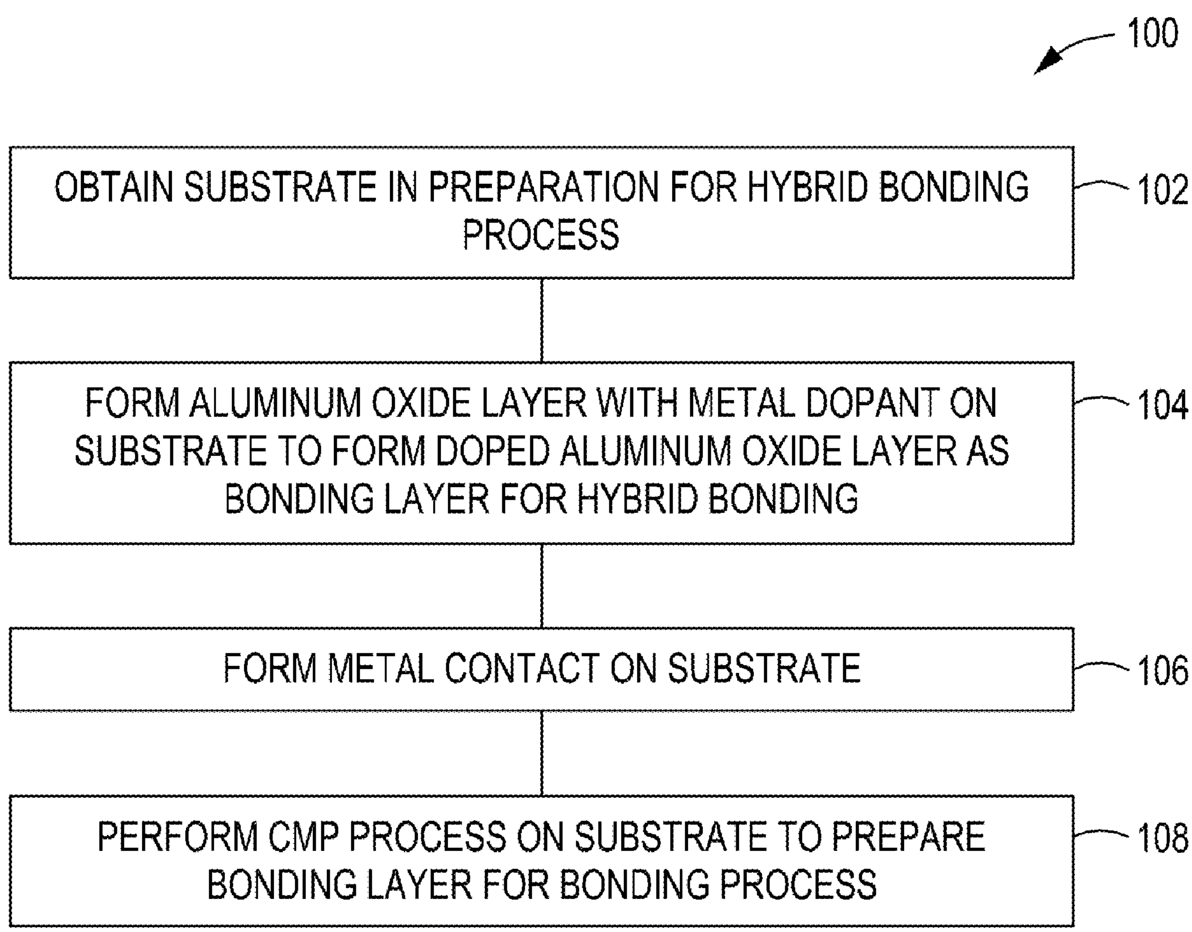
CPC ***H01L 21/02178*** (2013.01); ***C23C 16/403*** (2013.01); ***C23C 16/45525*** (2013.01); ***H01L 21/3215*** (2013.01)

(57) **ABSTRACT**

A method for processing a substrate for hybrid bonding that includes doping an aluminum oxide (Al_2O_3) layer prior to bonding of the substrate. A method may include forming an aluminum oxide layer with a metal dopant to form a doped aluminum oxide layer on the substrate where the doped aluminum oxide forms a bonding layer for a hybrid bonding process on an uppermost surface of the substrate. A metal contact is formed on the substrate that penetrates through the doped aluminum oxide layer and a chemical mechanical polish (CMP) process is performed to recess an uppermost surface of the metal contact below a surface of the doped aluminum oxide layer.



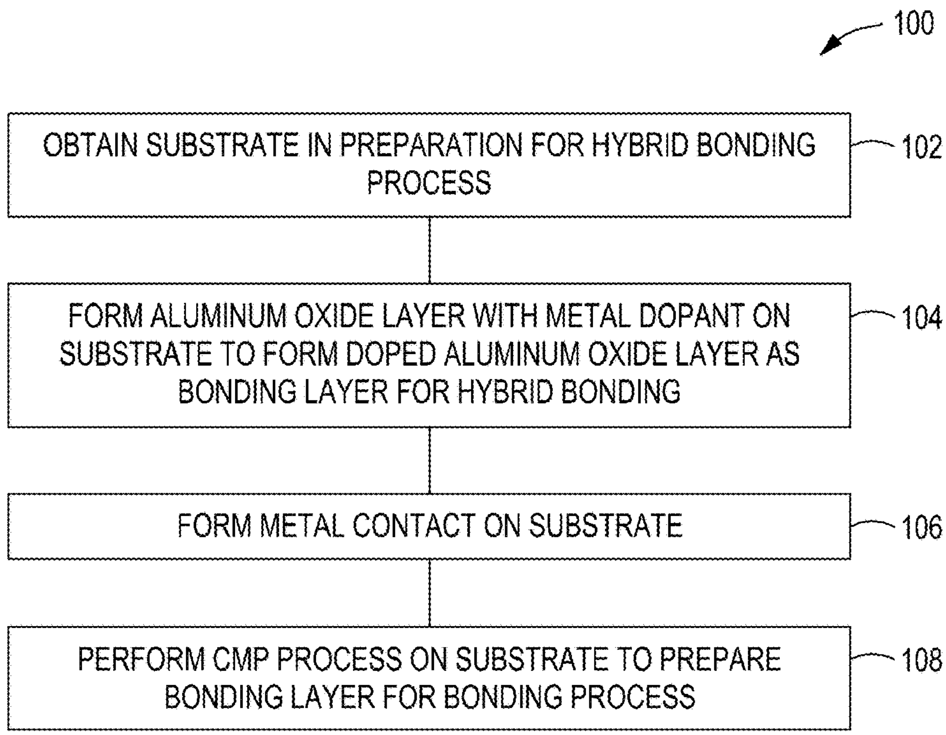

100
OBTAIN SUBSTRATE IN PREPARATION FOR HYBRID BONDING PROCESS
102
FORM ALUMINUM OXIDE LAYER WITH METAL DOPANT ON SUBSTRATE TO FORM DOPED ALUMINUM OXIDE LAYER AS BONDING LAYER FOR HYBRID BONDING
104
FORM METAL CONTACT ON SUBSTRATE
106
PERFORM CMP PROCESS ON SUBSTRATE TO PREPARE BONDING LAYER FOR BONDING PROCESS
108

FIG. 1

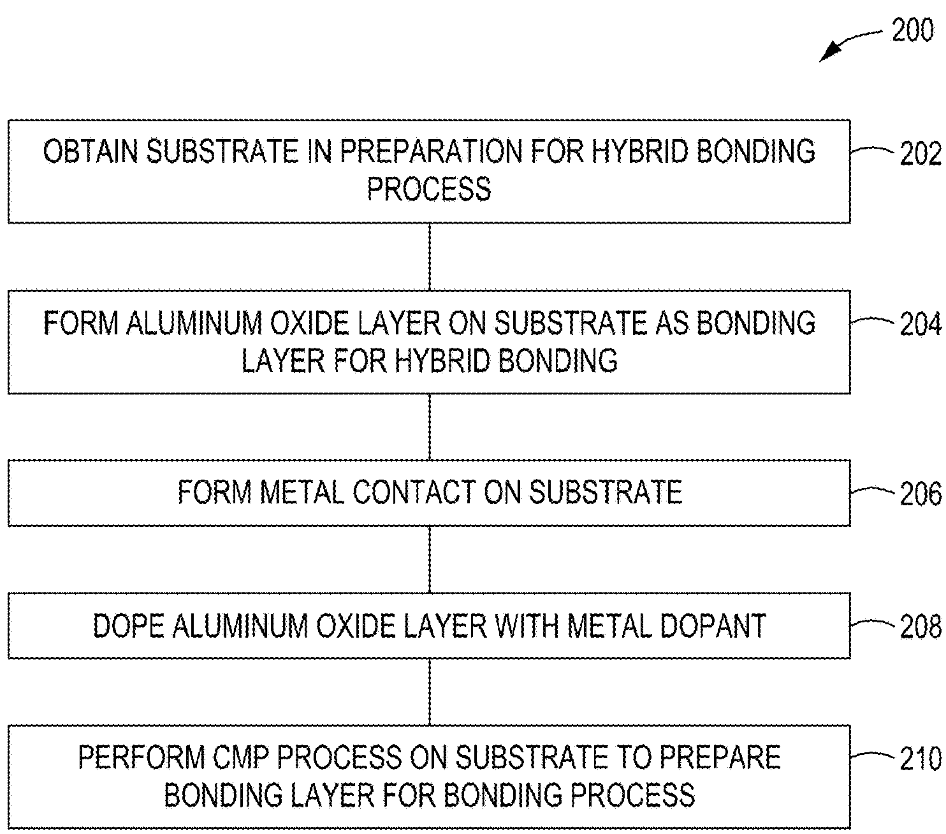

200
OBTAIN SUBSTRATE IN PREPARATION FOR HYBRID BONDING PROCESS
202
FORM ALUMINUM OXIDE LAYER ON SUBSTRATE AS BONDING LAYER FOR HYBRID BONDING
204
FORM METAL CONTACT ON SUBSTRATE
206
DOPE ALUMINUM OXIDE LAYER WITH METAL DOPANT
208
PERFORM CMP PROCESS ON SUBSTRATE TO PREPARE BONDING LAYER FOR BONDING PROCESS
210

FIG. 2

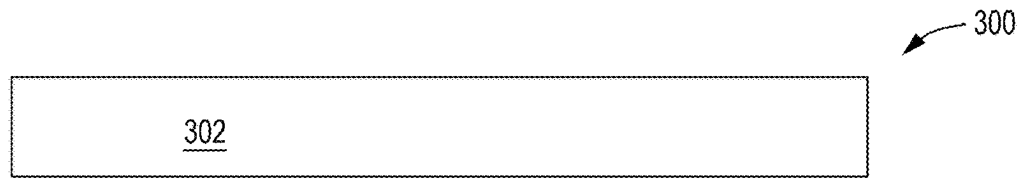


FIG. 3

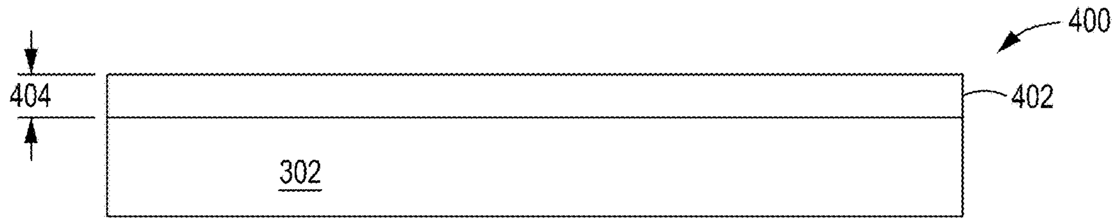


FIG. 4

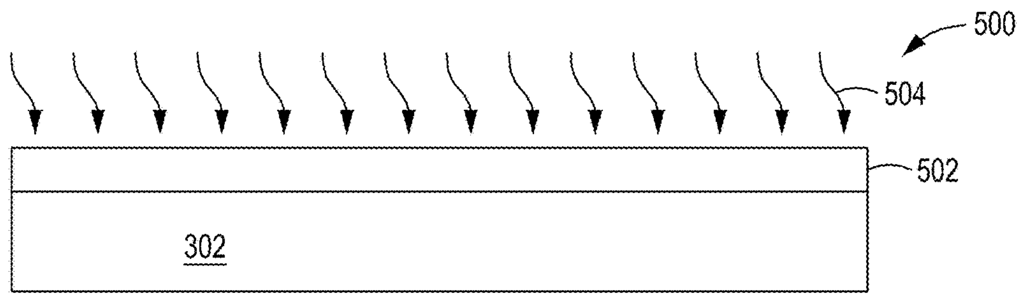


FIG. 5

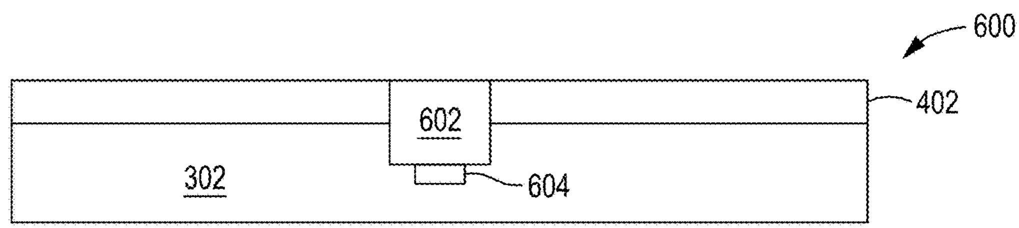


FIG. 6

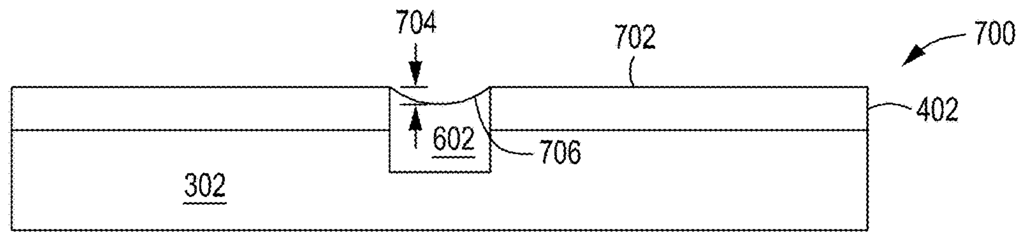


FIG. 7

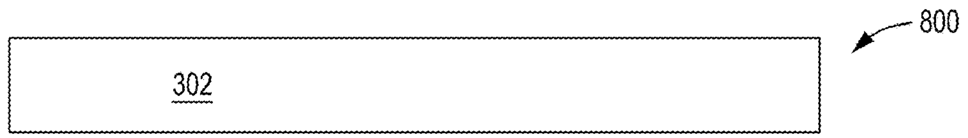


FIG. 8

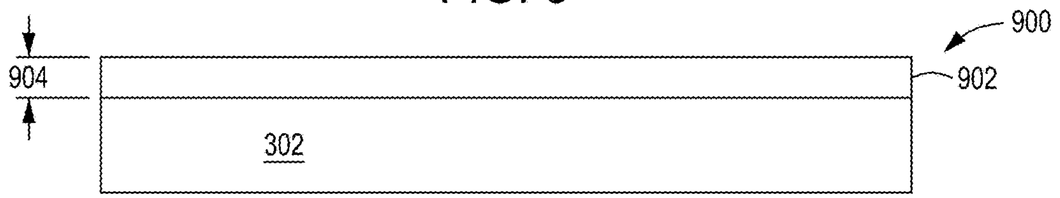


FIG. 9

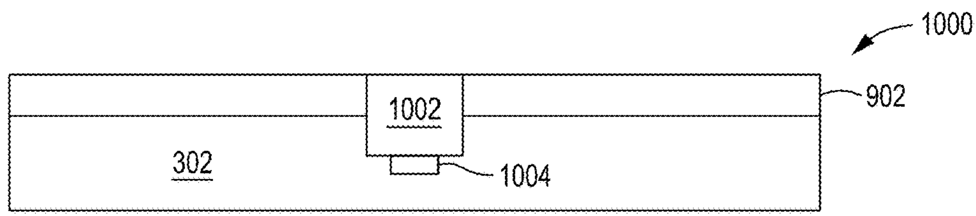


FIG. 10

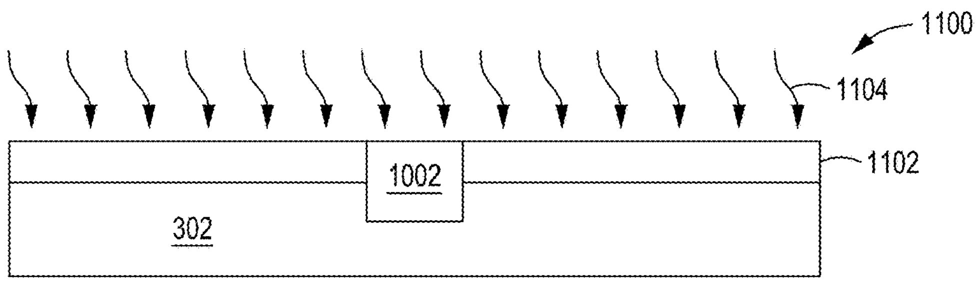


FIG. 11

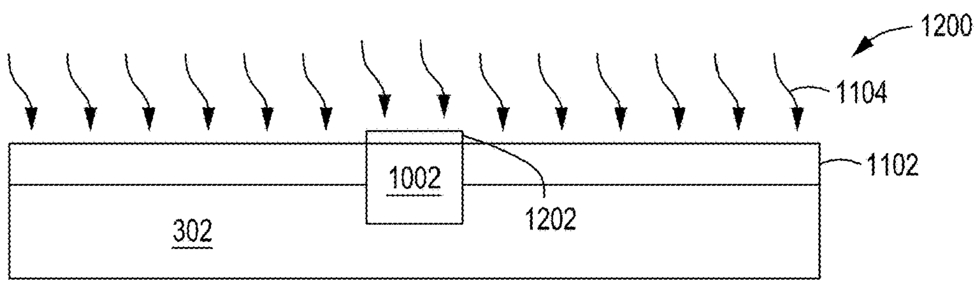


FIG. 12

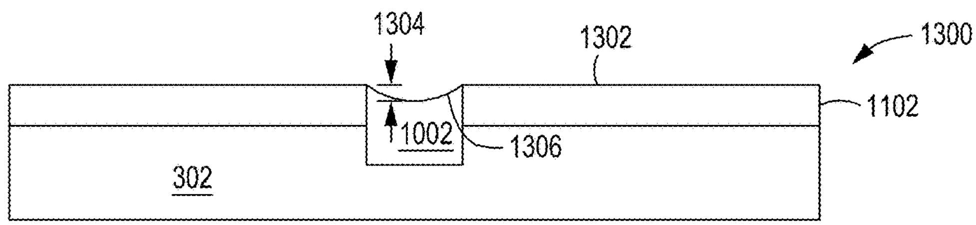


FIG. 13

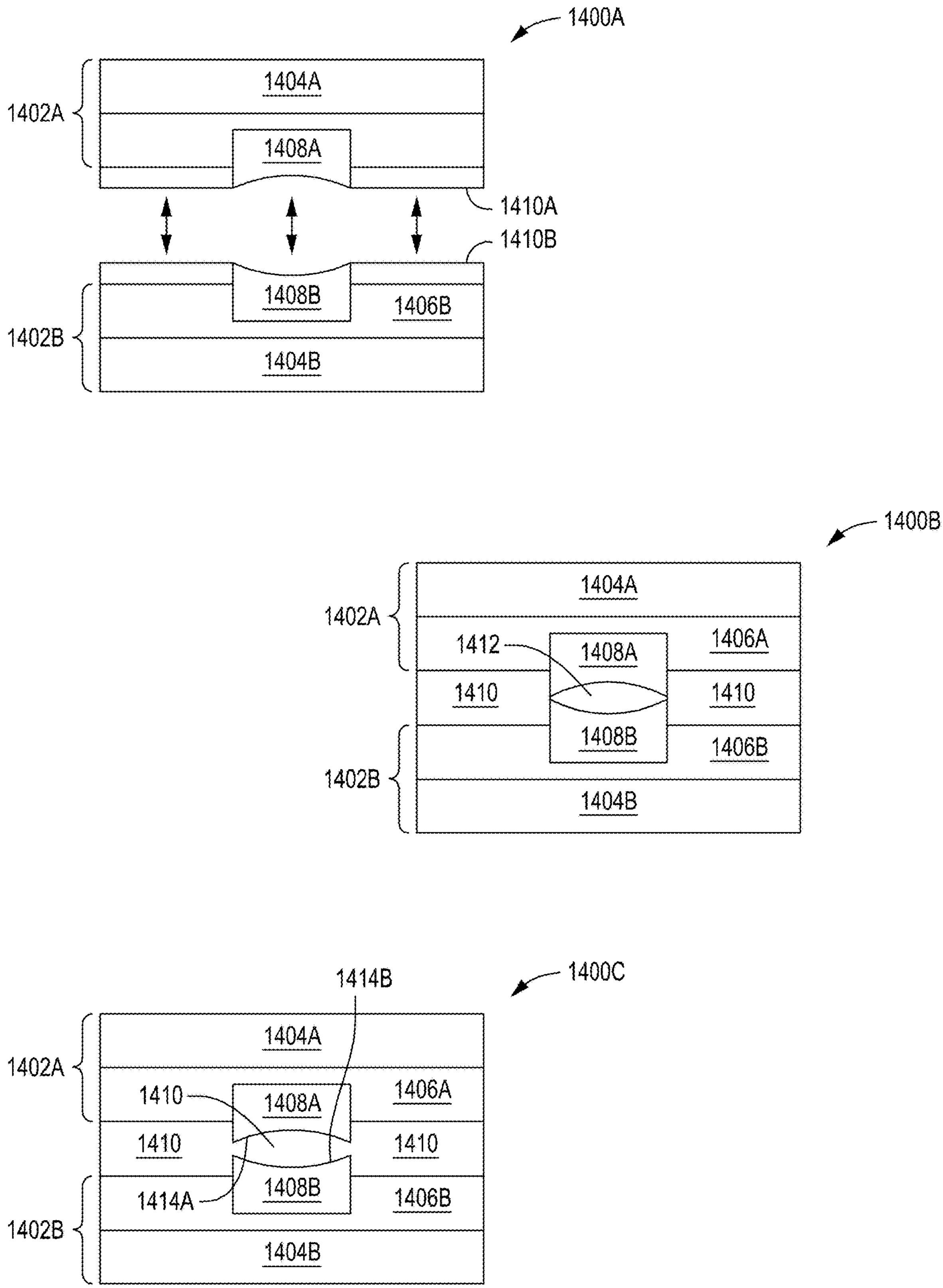

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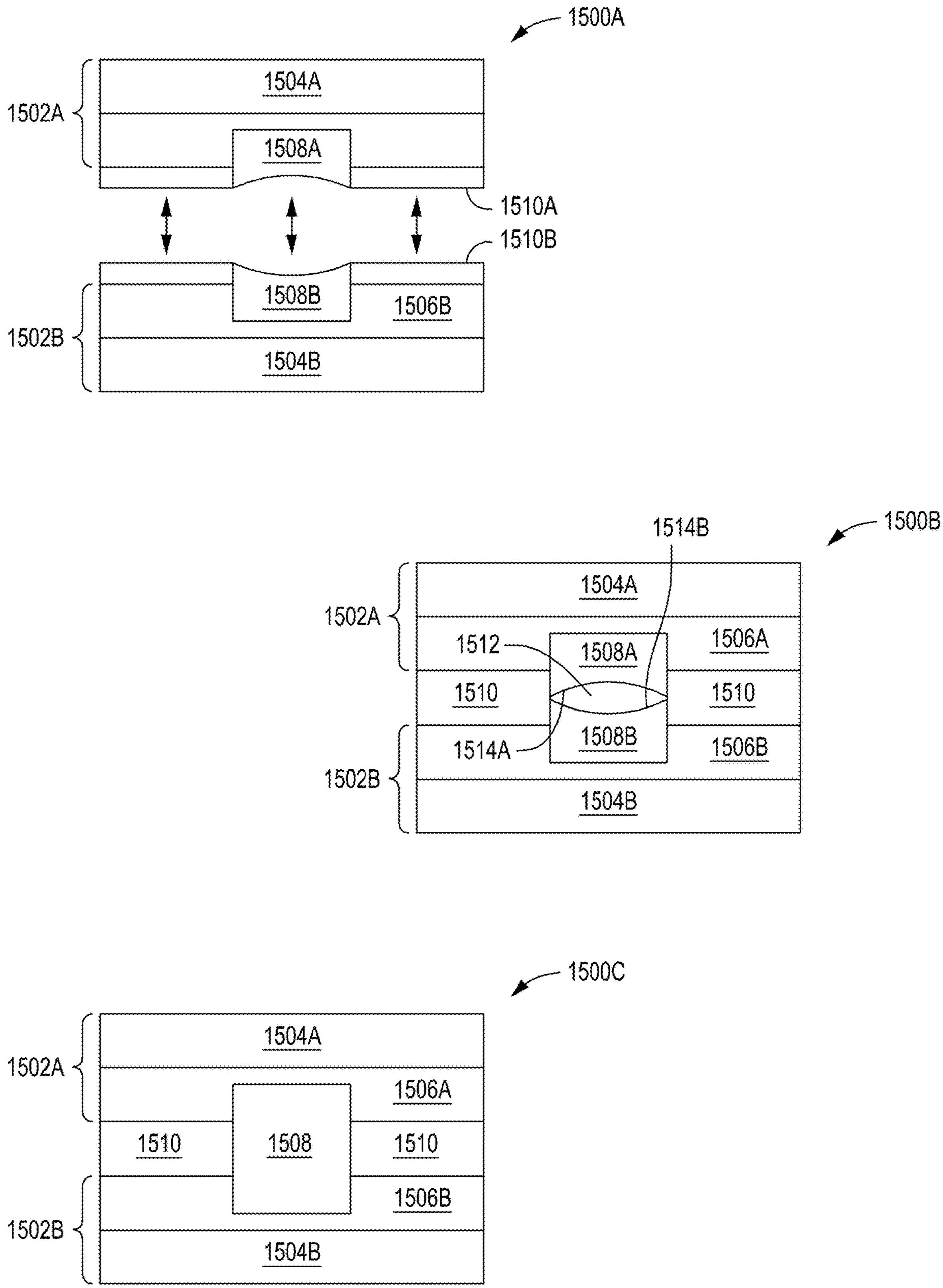
FIG. 14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1514A
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1506B
1502B
1504B
1500C
1504A
1502A
1506A
1510
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FIG. 15

DOPED ALUMINUM OXIDE BONDING LAYER FOR HYBRID BONDING

FIELD

[0001] Embodiments of the present principles generally relate to semiconductor processing of semiconductor substrates.

BACKGROUND

[0002] Hybrid bonding is the joining or bonding of more than one type of material. In semiconductor manufacturing, hybrid bonding is used to package devices by bonding one substrate or die to another substrate or die to join not only dielectric surfaces but to also join metal interconnects or contacts together. Traditionally, dielectrics, such as silicon dioxide, provide most of the bonding strength in a hybrid bond. As the density of the interconnects and contacts increases, the dielectric material bonding area decreases. To help ensure a successful bond with a die or substrate with a high density of contacts, dielectric materials with higher bonding strengths, such as aluminum oxide, may be introduced into the manufacturing processes. However, the inventors have observed that during hybrid bonding processes that use aluminum oxide, poor bonding occurs between the metal interconnects and contacts, leading to low yields.

[0003] Accordingly, the inventors have provided methods for improving hybrid bonding performance of aluminum oxide-based substrates.

SUMMARY

[0004] Methods for improving hybrid bonding performance by doping of aluminum oxides are provided herein.

[0005] In some embodiments, a method for processing a substrate may comprise providing a first substrate for a hybrid bonding process and forming, on the first substrate, a hybrid bonding layer that comprises a doped aluminum oxide (Al_2O_3) layer and a metal contact penetrating through the doped aluminum oxide layer.

[0006] In some embodiments, the method may further include hybrid bonding the first substrate to a second substrate via the hybrid bonding layer, a doped aluminum oxide layer that is doped with a metal dopant that has an atomic radius greater than an atomic radius of aluminum or oxygen, formation of a doped aluminum oxide layer that includes depositing an aluminum oxide layer in conjunction with the metal dopant, aluminum oxide and metal dopant that are co-sputtered using a physical vapor deposition (PVD) process to form the doped aluminum oxide layer with the metal dopant interspersed throughout, aluminum oxide and metal dopant that are deposited using an atomic layer deposition (ALD) process to form the doped aluminum oxide layer with the metal dopant interspersed throughout, metal dopant that is hafnium, zirconium, titanium, tungsten, or tantalum, an amount of the metal dopant in the doped aluminum oxide layer that is selected based on a temperature of an annealing process to be performed after a bonding process, a doped aluminum oxide layer that is an aluminum oxide layer that is formed first on the first substrate followed by a formation of the metal contact on the first substrate which penetrates through the aluminum oxide layer and then doped with a metal dopant throughout, doping of the aluminum oxide layer that is performed selectively only on the aluminum oxide layer, doping of the aluminum oxide layer that is performed on the aluminum oxide layer and on the metal contact, a doped aluminum oxide layer that is approximately 30 nm to approximately 50 nm in thickness, a doped aluminum oxide layer that has a leakage current of less than approximately 2× a leakage current of aluminum oxide or silicon dioxide, and/or a doped aluminum oxide layer that contains at least approximately 25% metal dopant.

[0007] In some embodiments, a substrate prepared for hybrid bonding may comprise a dielectric material, at least one metal contact, and a doped dielectric bonding layer surrounding the at least one metal contact where an uppermost surface of the at least one metal contact is exposed through the doped dielectric bonding layer.

[0008] In some embodiments, the substrate may further include a doped dielectric bonding layer that is a doped aluminum oxide (Al_2O_3) layer where at least one of the at least one metal contact is copper and where the doped aluminum oxide layer is doped with a metal dopant throughout, a metal dopant that is hafnium, zirconium, titanium, tungsten, or tantalum, and/or a device comprising the substrate another substrate that is hybrid bonded to the substrate via the doped dielectric bonding layer and the at least one metal contact.

[0009] In some embodiments, a non-transitory, computer readable medium having instructions stored thereon that, when executed, cause a method for processing a substrate, the method may comprise providing a substrate for a hybrid bonding process and forming, on the substrate, a hybrid bonding layer that comprises a doped aluminum oxide (Al_2O_3) layer and a metal contact penetrating through the doped aluminum oxide layer.

[0010] In some embodiments, the method of the non-transitory, computer readable medium may further include a doped aluminum oxide layer that has a leakage current of less than approximately 2× a leakage current of aluminum oxide or silicon dioxide, a doped aluminum oxide layer that is doped with a metal dopant and where an amount of the metal dopant in the doped aluminum oxide layer is selected based on a temperature of an annealing process to be performed after a bonding process, formation of a doped aluminum oxide layer that includes depositing an aluminum oxide layer on the substrate and doping the aluminum oxide layer with a metal dopant and/or formation of the doped aluminum oxide layer includes depositing an aluminum oxide layer in conjunction with a metal dopant.

[0011] Other and further embodiments are disclosed below.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Embodiments of the present principles, briefly summarized above and discussed in greater detail below, can be understood by reference to the illustrative embodiments of the principles depicted in the appended drawings. However, the appended drawings illustrate only typical embodiments of the principles and are thus not to be considered limiting of scope, for the principles may admit to other equally effective embodiments.

[0013] FIG. **1** is a method for preparing a substrate for hybrid bonding where metal contact formation occurs after doping procedures in accordance with some embodiments of the present principles.

[0014] FIG. **2** is a method for preparing a substrate for hybrid bonding where metal contact formation occurs before doping procedures in accordance with some embodiments of the present principles.

[0015] FIG. **3** depicts a cross-sectional view of a substrate in preparation for a hybrid bonding process in accordance with some embodiments of the present principles.

[0016] FIG. **4** depicts a cross-sectional view of a substrate with a doped aluminum oxide bonding layer in accordance with some embodiments of the present principles.

[0017] FIG. **5** depicts a cross-sectional view of an aluminum oxide bonding layer being doped with a metal dopant in accordance with some embodiments of the present principles.

[0018] FIG. **6** depicts a cross-sectional view of a substrate with a metal contact formed through a doped aluminum oxide bonding layer in accordance with some embodiments of the present principles.

[0019] FIG. **7** depicts a cross-sectional view of a substrate with a doped aluminum oxide bonding layer after a chemical mechanical planarization process in accordance with some embodiments of the present principles.

[0020] FIG. **8** depicts a cross-sectional view of a substrate in preparation for a hybrid bonding process in accordance with some embodiments of the present principles.

[0021] FIG. **9** depicts a cross-sectional view of a substrate with an aluminum oxide bonding layer in accordance with some embodiments of the present principles.

[0022] FIG. **10** depicts a cross-sectional view of a metal contact formed through an aluminum oxide bonding layer in accordance with some embodiments of the present principles.

[0023] FIG. **11** depicts a cross-sectional view of an aluminum oxide bonding layer being doped with a metal dopant in a global doping process in accordance with some embodiments of the present principles.

[0024] FIG. **12** depicts a cross-sectional view of a selective doping process of an aluminum oxide bonding layer in accordance with some embodiments of the present principles.

[0025] FIG. **13** depicts a cross-sectional view of a substrate with a doped aluminum oxide bonding layer after a chemical mechanical planarization process in accordance with some embodiments of the present principles.

[0026] FIG. **14** depicts cross-sectional views of a hybrid bonding process without a doped aluminum oxide bonding layer of the present principles.

[0027] FIG. **15** depicts cross-sectional views of a hybrid bonding process using a doped aluminum oxide bonding layer in accordance with some embodiments of the present principles.

[0028] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. The figures are not drawn to scale and may be simplified for clarity. Elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0029] The methods provide improved hybrid bonding performance by doping aluminum oxides deposited as a bonding surface. The doped aluminum oxide layer prevents migration of aluminum oxide over copper contacts during annealing of the bonded substrates, providing both high dielectric and contact bonding strengths. The high bonding strengths provided by the present methods enable increased metal contact densities to be achieved, allowing for scaling of semiconductor devices. The present techniques may be performed prior to contact formation on a substrate or after contact formation on the substrate, providing processing flexibility and easy integration into existing hybrid bonding preparation processes.

[0030] Traditional dielectrics such as silicon dioxide limit the scaling of devices due to the bonding strength of the dielectric material. In order to increase the metal contact densities, the bonding strength of the dielectrics used in hybrid bonding must be increased as the bonding strength of the metal contacts is relatively low compared to the dielectric bonding strength. Aluminum oxide is being considered as a replacement for dielectric materials such as silicon dioxide. Aluminum oxide has a 40% to 80% increase in bonding strength compared to that of silicon dioxide. However, the inventors have found that when a substrate with aluminum oxide and metal contacts are bonded together and then annealed, the aluminum oxide migrates over the top surfaces of the metal contacts and prevents the metal contacts from bonding. The inventors believe that the migration of the aluminum oxide is caused by low-temperature crystallization of the aluminum oxide. Typically, aluminum oxide crystallizes at a temperature of approximately 850 degrees Celsius. Hybrid bonding annealing temperatures are generally around 350 degrees Celsius. The inventors believe that the crystals of the metal material (e.g., copper, etc.) of the metal contacts acts as a catalyst for the crystallization of the aluminum oxide which lowers the crystallization temperature to at or below the annealing temperature. The low temperature causes the aluminum oxide to crystallize and migrate/diffuse over the surfaces of the metal contacts during annealing of the bonded substrate (e.g., the aluminum oxide has a higher diffusion rate than copper at a given temperature causing aluminum oxide to migrate).

[0031] The inventors found that to successfully bond using an aluminum oxide bonding layer, the aluminum oxide crystallization and subsequent migration need to be prevented. Suppression of the aluminum oxide crystallization during hybrid bonding may be accomplished by doping the aluminum oxide with a crystal-suppressing element or by disrupting the aluminum oxide crystal lattice. Any element that suppresses crystallization of aluminum oxide is suitable. For example, elements that do not easily fit in the lattice points of aluminum or oxygen in aluminum oxide, such as, but not limited to, hafnium and the like may be used to suppress aluminum oxide crystallization. The inventors found that by doping the aluminum oxide with such an element, the migration of the aluminum oxide does not occur during hybrid bonding anneal processes. The dopant can be implanted on the entire substrate, including into surfaces of the metal contacts, or selective only into the aluminum oxide, as long as the dopant does not make the aluminum oxide conductive or interfere with the metal contact expansion and bonding. Alternatively, the dopant can be deposited in conjunction with the aluminum oxide to form a doped aluminum oxide in a single deposition process.

[0032] The doping may be performed by using a plasma treatment process, a thermal treatment process, an ion bombardment process, an alloy deposition process (aluminum oxide plus dopant deposited at the same time) (e.g., ALD or co-sputtering, etc.), and the like. Dopants may include elements with an atomic radius much greater than aluminum or oxygen (such as, e.g., Hf, Zr, Ti, W, Ta) or much smaller than aluminum or oxygen to disrupt the crystal lattice by altering the size or shape of bonds in the region. Dopants that include elements which form different crystal structures than aluminum oxide (Al_2O_3) may also be used for crystallization suppress of the aluminum oxide. The aluminum oxide doping suppresses the grain nucleation (location at which crystal growth begins) by using elements that disrupt the aluminum crystal lattice to prevent the crystallization at low temperatures (approximately 300 degrees Celsius to approximately 400 degrees Celsius). For example, monoclinic lattice structures can be used to disrupt the crystal lattice structure of the aluminum oxide. Adjustments can be made to the dopant concentration and the annealing temperature to produce a doped aluminum oxide bonding layer with suppressed crystallization and migration characteristics. Having the metal dopant interspersed throughout the aluminum oxide layer is preferred as subsequent chemical mechanical planarization (CMP) processes may remove some portions of the aluminum oxide layer. If doping is shallow, a significant portion of the doping may be removed during the planarization process, reducing the crystallization suppression performance of the doped aluminum oxide.

[0033] FIG. **1** is a method **100** for preparing a substrate **302** for hybrid bonding. References will be made to FIGS. **3-7** during the discussion of the method **100**. In block **102**, a substrate is obtained or provided in preparation for the hybrid bonding process as depicted in a view **300** of FIG. **3**. The substrate **302** may be formed, at least in part, of dielectric materials such as, for example but not limited to, silicon dioxide and/or silicon carbon nitride materials and the like. In some embodiments, the substrate **302** may have undergone prior device manufacturing and/or packaging processes to prepare the substrate **302** for bonding to another substrate and/or die. In block **104**, an aluminum oxide layer with a metal dopant is formed on the substrate **302** to form a doped aluminum oxide layer **402** as a bonding layer for a hybrid bonding process as depicted in a view **400** of FIG. **4**. The doped aluminum oxide layer **402** may have a thickness **404** of approximately 30 nm to approximately 50 nm.

[0034] In some embodiments, the amount of dopant and processes used for doping are selected to ensure that the leakage current of the doped aluminum oxide layer **402** is less than approximately 2× (two times) that of aluminum oxide or silicon dioxide while still suppressing the crystallization of the aluminum oxide. The metal dopant is conductive in nature to varying degrees, and the conductivity of the metal dopant is taken into account to determine the amount of doping that ensures suppression of crystallization of the aluminum oxide while also ensuring that the doped aluminum oxide layer **402** still serves as an insulative dielectric on the surface of the substrate **302**. In some embodiments, the leakage current of the doped aluminum oxide layer **402** may be approximately 100 nA/cm^2 or less.

[0035] Formation of the doped aluminum oxide layer **402** prior to formation of the metal contacts on the substrate **302** allows the doping process to use oxygen during the doping process without forming oxides on the metal contacts. The oxygen can be used to control the leakage current of the doped aluminum oxide layer **402** by mitigating the conductivity of the metal dopant as the doped aluminum oxide layer **402** is formed. Oxygenating the doping process minimizes the leakage current issue significantly, as the aluminum oxide will remain an insulator after doping. Without having to control the leakage current, the sole focus becomes suppression of the crystallization of the aluminum oxide, allowing higher concentrations of the metal dopant to be used to ensure that the migration of the aluminum oxide does not occur during annealing (e.g., 50% metal dopant (or higher) and 50% aluminum oxide (or less), etc.).

[0036] As the metal contacts on the substrate **302** promote crystallization and migration of the aluminum oxide at much lower temperatures (aluminum oxide crystallizes at 850 degrees Celsius and inventors have observed crystallization of aluminum oxide at less than 400 degrees Celsius during hybrid bonding annealing), the amount of metal dopant may be selected based on the temperature of the annealing processes that are performed after the bonding process of a hybrid bonding procedure. In some embodiments, the doped aluminum oxide layer may contain at least approximately 25% metal dopant to prevent crystallization of the aluminum oxide. In some embodiments, the doped aluminum oxide layer may contain at least approximately 50% metal dopant. As higher temperatures promote crystallization, annealing temperatures of approximately 300 degrees Celsius may require lower amounts of metal dopants and annealing temperatures of approximately 350 degrees or higher may require higher amounts of dopants to prevent crystallization of the aluminum oxide. The amount of metal dopant used is based on the temperatures used in the hybrid bonding annealing procedures and the leakage current requirements (where leak current is applicable-oxygenated (not applicable), nonoxygenated (applicable) doping processes) to keep the doped aluminum oxide layer **402** insulative.

[0037] Some metal dopants have higher conductivity than other metal dopants and drive the leakage current higher with lower amounts of dopant. As such, higher conductivity may cause too high of a leakage current at a given amount of dopant which is not sufficient to suppress the crystallization of the aluminum oxide at a given annealing temperature. Thus, the type of dopant, due to the dopant's conductivity, may need to be altered based on the leakage and crystallization suppression characteristics for a given annealing temperature. In general, the higher the metal dopant concentration, the higher the crystallization temperature of the aluminum oxide. However, the higher metal dopant concentrations are limited by the acceptable leakage current characteristics of a given application (which can change based on the application). Formation of the doped aluminum oxide layer using oxygen can eliminate the leakage current from the doping equation.

[0038] Aluminum oxide has a 40% to 80% increased bonding strength over the silicon dioxide of the substrate **302**. An aluminum oxide layer would provide a substantial improvement when used as a bonding layer if not for migration of the aluminum oxide during subsequent annealing processes of a hybrid bonding process. The present methods incorporate the doped aluminum oxide layer **402** to overcome the deficiencies of the aluminum oxide while providing all or at least a substantial portion of the increased bonding strength of the aluminum oxide over silicon dioxide and the like. In some embodiments, the doped aluminum oxide layer **402** is formed by co-sputtering aluminum oxide in conjunction with the metal dopant (and, optionally, oxygen) such that the metal dopant is interspersed throughout the doped aluminum oxide layer **402**. In some embodiments, the doped aluminum oxide layer **402** is formed by an atomic layer deposition (ALD) process that includes precursors for the aluminum oxide layer formation in conjunction with precursors for the metal dopant (and, optionally, oxygen) such that the metal dopant is interspersed through the doped aluminum oxide layer **402**.

[0039] In some embodiments, in the alternative, an aluminum oxide layer **502** is first deposited on the substrate **302** and then a doping process **504** is used to dope the aluminum oxide layer **502** to form the doped aluminum oxide layer **402** as depicted in a view **500** of FIG. **5**. The aluminum oxide layer may be doped using a plasma-based doping process, a thermal-based doping process, or an ion bombardment-based doping process, and the like. Oxygen may be introduced during the doping process. As discussed above, the metal dopant has a metal dopant atomic radius greater than an atomic radius of aluminum or oxygen to facilitate in suppressing crystallization of the aluminum oxide in the doped aluminum oxide layer **402**. In some embodiments, the metal dopant may be hafnium, zirconium, titanium, tungsten, or tantalum, and the like. The size of the metal dopant helps to suppress the formation of the crystallization of the aluminum oxide during annealing which prevents the migration of the aluminum oxide over the metal contacts.

[0040] In block **106**, a metal contact **602** is formed into the doped aluminum oxide layer **402** as depicted in a view **600** of FIG. **6**. The metal contact **602** penetrates through the doped aluminum oxide layer **402** to make contact with underlying interconnects **604** and the like. The doped aluminum oxide layer **402** and the metal contact **602** form a hybrid bonding layer. In block **108**, a CMP process is performed on the substrate **302** as depicted in a view **700** of FIG. **7**. The CMP process planarizes the uppermost surface **702** of the substrate **302** and is configured to recess **704** or dish the uppermost surface **706** of the metal contact **602** below the uppermost surface **702** of the doped aluminum oxide layer **402**. The recess **704** of the metal contact **602** allows for expansion of the metal contact **602** during annealing processes of the hybrid bonding process. The substrate **302** is then bonded to a similarly prepared substrate and/or die by bringing the uppermost surface **702** of the substrate **302** into contact with another doped aluminum oxide layer on another substrate or die. The bonded substrates are then annealed to further increase the bonding strength of the doped aluminum oxide layers and to bond the metal contacts to one another. The annealing process typically is performed at a temperature of approximately 300 degrees Celsius to approximately 400 degrees Celsius. As the bonding process is a back end of line (BEOL) process, the semiconductor devices on the substrate have a thermal budget that typically cannot substantially exceed 400 degrees Celsius.

[0041] In some instances, the order in which the doping of the substrate **302** occurs may be altered. FIG. **2** is a method **200** for preparing the substrate **302** for hybrid bonding. References will be made to FIGS. **8-13** during the discussion of the method **200**. In block **202**, the substrate **302** is obtained in preparation for the hybrid bonding process as depicted in a view **800** of FIG. **8**. The substrate **302** may be formed, at least in part, of silicon dioxide materials. In some embodiments, the substrate **302** may have undergone prior device manufacturing and/or packaging processes to prepare the substrate **302** for bonding to another substrate and/or die. In block **204**, an aluminum oxide layer **902** is formed on the substrate **302** as a dielectric bonding layer for a hybrid bonding process as depicted in a view **900** of FIG. **9**. The aluminum oxide layer **902** may have a thickness **904** of approximately 30 nm to approximately 50 nm. In block **206**, a metal contact **1002** is formed into the aluminum oxide layer **902** as depicted in a view **1000** of FIG. **10**. The metal contact **1002** penetrates through the aluminum oxide layer **902** to make contact with underlying interconnects **1004** and the like.

[0042] In block **208**, the aluminum oxide layer **902** is doped **1104** with a metal dopant to form a doped aluminum oxide layer **1102** as depicted in a view **1100** of FIG. **11**. The doped aluminum oxide layer **1102** and the metal contact **1002** form a hybrid bonding layer. In some embodiments, the metal dopant is implanted in both the aluminum oxide layer **902** and into the metal contact **1002**. The metal dopant, being conductive, does not directly impact the conductivity of the metal contact **1002**. Care must be taken to avoid affecting the bonding of the metal contact **1002** during the annealing process of the hybrid bonding procedure. The amount of metal dopant is typically insufficient to affect bonding performance of the metal contact **1002**, as the metal material of the metal contact (e.g., copper and the like) still has sufficiently large numbers of atoms that far exceed the dopant amounts. The advantage of globally doping the surface of the substrate **302** is that the process is quicker and easier to accomplish than with selective doping processes discussed next. The disadvantage is that oxygen cannot be introduced in the doping process to control the leakage current due to oxides forming on the exposed metal contacts.

[0043] In some embodiments, as depicted in the view **1200** of FIG. **12**, the metal dopant is implanted only in the aluminum oxide layer **902** and not in the metal contact **1002**. The selective doping may be accomplished through photolithography and the like. A photoresist layer **1202** may be patterned on the surface of the metal contact **1002** to prevent doping of the metal contact **1002**. Selective doping avoids the issues as discussed above for global doping but requires extra processing for the photolithography and the like. The advantage of selective doping is reactions to the dopant in the metal contact **1002** is not an issue and, therefore, does not impact the type or quantity of dopant that may occur due to the dopant's influence on the bonding properties of the metal contact **1002**. In addition, oxygen can be introduced during the doping process to eliminate the leakage current of the aluminum oxide as the metal contact **1002** is not exposed to the oxygen, which would oxidize the metal material of the metal contact **1002** and negatively impact the metal contact's conductivity.

[0044] In block **210**, similar to block **108** of method **100**, a CMP process is performed on the substrate **302** as depicted in a view **1300** of FIG. **13**. The CMP process planarizes the uppermost surface **1302** of the substrate **302** and is configured to recess **1304** or dish the uppermost surface **1306** of the metal contact **1002** below the uppermost surface **1302** of the doped aluminum oxide layer **1102**. The recess **1304** of the metal contact **1002** allows for expansion of the metal contact **1002** during annealing processes of the hybrid bonding process. The substrate **302** is then bonded to a similarly prepared substrate and/or die (by method **100** or by method **200** and similar) by bringing the uppermost surface **1302** of the substrate **302** into contact with another doped aluminum oxide layer on another substrate or die. The bonded substrates are then annealed to further increase the bonding strength of the doped aluminum oxide layers and to bond the metal contacts to one another. The annealing process typically is performed at a temperature of approximately 300 degrees Celsius to approximately 400 degrees Celsius. As the bonding process is a BEOL process, the semiconductor devices on the substrate have a thermal budget that typically cannot substantially exceed 400 degrees Celsius.

[0045] The above methods can be used to enhance the bonding strength and yield of hybrid bonding processes. For example, as depicted in a view **1400**A of FIG. **14**, the substrates **1402**A, **1402**B may be bonded together. The substrates **1402**A, **1402**B have silicon layers **1404**A, **1404**B, silicon dioxide layers **1406**A, **1406**B, and metal contacts **1408**A, **1408**B. The bonding layers are aluminum oxide layers **1410**A, **1410**B. As depicted in a view **1400**B of FIG. **14**, when the substrates **1402**A, **1402**B with aluminum oxide layers **1410**A, **1410**B are brought into contact with each other (bonded), the aluminum oxide layers **1410**A, **1410**B form a single aluminum oxide bonded layer **1410**. The metal contacts **1408**A, **1408**B have not yet bonded together, and a gap **1412** exists between the metal contacts **1408**A, **1408**B due to the recesses formed on each metal contact during CMP processing. In order to close the gap **1412** and form a continuous contact, the substrates **1402**A, **1402**B undergo an annealing process at temperatures up to 400 degrees Celsius.

[0046] The inventors have found that during the annealing process, the aluminum oxide material of the aluminum oxide layers crystallize at a dramatically lower temperature (at 400 degrees Celsius or less compared to a normal crystallization temperature of 850 degrees Celsius) due to the material (e.g., copper, etc.) of the metal contacts acting as a crystallization catalyst. As depicted in a view **1400**C of FIG. **14**, the substantially lowered crystallization temperature allows the single aluminum oxide bonded layer **1410** to crystallize and migrate over the surfaces **1414**A, **1414**B of the metal contacts **1408**A, **1408**B during the annealing process, preventing the metal contacts **1408**A, **1408**B from bonding together.

[0047] As depicted in a view **1500**A of FIG. **15**, the substrates **1502**A, **1502**B may be bonded together using the present methods as described above. In some embodiments, the substrates **1502**A, **1502**B may include dielectric layers such as, but not limited to, silicon dioxide and/or silicon carbon nitride and the like. In the example, the substrates **1502**A, **1502**B have silicon layers **1504**A, **1504**B, silicon dioxide layers **1506**A, **1506**B, and metal contacts **1508**A, **1508**B. The bonding layers are doped aluminum oxide layers **1510**A, **1510**B. As depicted in a view **1500**B of FIG. **15**, when the substrates **1502**A, **1502**B with doped aluminum oxide layers **1510**A, **1510**B are brought into contact with each other (bonded), the doped aluminum oxide layers **1510**A, **1510**B form a single doped aluminum oxide bonded layer **1510**. The metal contacts **1508**A, **1508**B have not yet bonded together, and a gap **1512** exists between the metal contacts **1508**A, **1508**B due to the recesses formed on each metal contact during CMP processing. In order to close the gap **1512** and form a continuous contact, the substrates **1502**A, **1502**B undergo an annealing process at temperatures up to approximately 400 degrees Celsius.

[0048] The inventors discovered that by doping the aluminum oxide prior to the annealing process, the aluminum oxide material of the doped aluminum oxide layers did not crystallize at the lower annealing temperatures (at approximately 300 degrees Celsius to approximately 400 degrees Celsius) in spite of the contact material (e.g., copper, etc.) of the metal contacts acting as a crystallization catalyst. As depicted in a view **1500**C of FIG. **15**, the single doped aluminum oxide bonded layer **1510** does not crystallize and migrate over the surfaces **1514**A, **1514**B of the metal contacts **1508**A, **1508**B during the annealing process, allowing the metal contacts **1508**A, **1508**B to bond together to form a continuous interconnect **1508**.

[0049] Embodiments in accordance with the present principles may be implemented in hardware, firmware, software, or any combination thereof. Embodiments may also be implemented as instructions stored using one or more computer readable media, which may be read and executed by one or more processors. A computer readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computing platform or a "virtual machine" running on one or more computing platforms). For example, a computer readable medium may include any suitable form of volatile or non-volatile memory. In some embodiments, the computer readable media may include a non-transitory computer readable medium.

[0050] While the foregoing is directed to embodiments of the present principles, other and further embodiments of the principles may be devised without departing from the basic scope thereof.

1. A method for processing a substrate, comprising:

providing a first substrate for a hybrid bonding process; and

forming, on the first substrate, a hybrid bonding layer that comprises:

a doped aluminum oxide (Al_2O_3) layer; and

a metal contact penetrating through the doped aluminum oxide layer.

2. The method of claim **1**, further comprising:

hybrid bonding the first substrate to a second substrate via the hybrid bonding layer.

3. The method of claim **1**, wherein the doped aluminum oxide layer is doped with a metal dopant that has an atomic radius greater than an atomic radius of aluminum or oxygen.

4. The method of claim **3**, wherein formation of the doped aluminum oxide layer includes depositing an aluminum oxide layer in conjunction with the metal dopant.

5. The method of claim **4**, wherein aluminum oxide and the metal dopant are co-sputtered using a physical vapor deposition (PVD) process to form the doped aluminum oxide layer with the metal dopant interspersed throughout.

6. The method of claim **4**, wherein aluminum oxide and the metal dopant are deposited using an atomic layer deposition (ALD) process to form the doped aluminum oxide layer with the metal dopant interspersed throughout.

7. The method of claim **3**, wherein the metal dopant is hafnium, zirconium, titanium, tungsten, or tantalum.

8. The method of claim **3**, wherein an amount of the metal dopant in the doped aluminum oxide layer is selected based on a temperature of an annealing process to be performed after a bonding process.

9. The method of claim **1**, wherein the doped aluminum oxide layer is an aluminum oxide layer that is formed first on the first substrate followed by a formation of the metal contact on the first substrate which penetrates through the aluminum oxide layer and then doped with a metal dopant throughout.

10. The method of claim **9**, wherein doping of the aluminum oxide layer is performed selectively only on the aluminum oxide layer.

11. The method of claim **9**, wherein doping of the aluminum oxide layer is performed on the aluminum oxide layer and on the metal contact.

12. The method of claim **1**, wherein the doped aluminum oxide layer is approximately 30 nm to approximately 50 nm in thickness.

13. The method of claim **1**, wherein the doped aluminum oxide layer has a leakage current of less than approximately 2× a leakage current of aluminum oxide or silicon dioxide.

14. The method of claim **1**, wherein the doped aluminum oxide layer contains at least approximately 25% metal dopant.

15. A substrate prepared for hybrid bonding, comprising:

a dielectric material;

at least one metal contact; and

a doped dielectric bonding layer surrounding the at least one metal contact, wherein an uppermost surface of the at least one metal contact is exposed through the doped dielectric bonding layer.

16. The substrate of claim **15**, wherein the doped dielectric bonding layer is a doped aluminum oxide (Al_2O_3) layer, wherein at least one of the at least one metal contact is copper, and wherein the doped aluminum oxide layer is doped with a metal dopant throughout.

17. The substrate of claim **16**, wherein the metal dopant is hafnium, zirconium, titanium, tungsten, or tantalum.

18. A device, comprising:

the substrate of claim **15**; and

another substrate that is hybrid bonded to the substrate via the doped dielectric bonding layer and the at least one metal contact.

19. A non-transitory, computer readable medium having instructions stored thereon that, when executed, cause a method for processing a substrate, the method comprising:

providing a substrate for a hybrid bonding process; and

forming, on the substrate, a hybrid bonding layer that comprises:

a doped aluminum oxide (Al_2O_3) layer; and

a metal contact penetrating through the doped aluminum oxide layer.

20. The non-transitory, computer readable medium of claim **19**, wherein the method further comprises (a), (b), (c), or (d):

a) wherein the doped aluminum oxide layer has a leakage current of less than approximately 2× a leakage current of aluminum oxide or silicon dioxide;

b) wherein the doped aluminum oxide layer is doped with a metal dopant and wherein an amount of the metal dopant in the doped aluminum oxide layer is selected based on a temperature of an annealing process to be performed after a bonding process;

c) wherein formation of the doped aluminum oxide layer includes depositing an aluminum oxide layer on the substrate and then doping the aluminum oxide layer with a metal dopant; or

d) wherein formation of the doped aluminum oxide layer includes depositing an aluminum oxide layer in conjunction with a metal dopant.

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